

Cyclotomic Polynomials

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1 Why do we care about Cyclotomic Polynomials?

1. Every finite Division Ring is in fact a field.
2. Weak Dirichlet Theorem on Primes in Arithmetic Progressions: for any integer n , there are infinitely many primes of the form $kn + 1$.
3. Zsigmondy's Theorem: for any integers $a > b > 0$ and $n > 1$, there exists a prime that divides $a^n - b^n$ but does not divide $a^k - b^k$ for any $k < n$, with some exceptions.

2 What is a Cyclotomic Polynomial?

1. Definition: The n -th cyclotomic polynomial $\Phi_n(x)$ is defined as the monic polynomial whose roots

are the primitive n -th roots of unity. Explicitly,

$$\Phi_n(x) = \prod_{\substack{1 \leq k \leq n \\ \gcd(k,n)=1}} (x - e^{2\pi i k/n}).$$

Example 2.1

$\Phi_p(x) = x^{p-1} + x^{p-2} + \cdots + x + 1$ for a prime p . $\Phi_1(x) = x - 1$. $\Phi_2(x) = x + 1$. $\Phi_3(x) = x^2 + x + 1$. $\Phi_4(x) = x^2 + 1$.

2. $x^n - 1$ can be factored as

$$x^n - 1 = \prod_{d|n} \Phi_d(x).$$

Why?

3. Its degree is $\phi(n)$.

4. $\Phi_n(x)$ is a monic polynomial with integer coefficients.

Proof. Use induction on n . $\Phi_n(x) \mid x^n - 1$ in $\mathbb{C}[x]$, its quotient is $f(x) = \prod_{d|n, d < n} \Phi_d(x)$, which is in $\mathbb{Z}[x]$ by inductive hypothesis. Consider the long division of $x^n - 1$ by $f(x)$ in $\mathbb{Z}[x]$. $x^n - 1 = f(x)q(x) + r(x)$. So $f(x)(q(x) - \Phi_n(x)) = -r(x)$. So $r(x) = 0$ and $q(x) = \Phi_n(x) \in \mathbb{Z}[x]$. $\square$

5. $\Phi_n(x)$ is irreducible in $\mathbb{Q}[x]$.

Proof. Let $f(x)$ be the irreducible proper factor of $\Phi_n(x)$.

Case 1: There exists m and $p \nmid n$ such that $f(\zeta_n^m) = 0$ but $f(\zeta_n^{pm}) \neq 0$.

Let $\Phi_n(x) = f(x)g(x)$. So ζ_n^m is a root of $g(x^p)$. Since f is irreducible, it must be the minimal polynomial of ζ_n^m . So $f(x) \mid g(x^p)$ in $\mathbb{Z}[x]$. Thus, $f(x) \mid g(x^p) = g(x)^p$ in $\mathbb{F}_p[x]$. So there exists an irreducible $l(x) \mid f(x) \mid g(x)^p$ so $l(x) \mid g(x)$. So $l(x)^2 \mid \Phi_n(x) \mid x^n - 1$. Since $p \nmid n$, $x^n - 1$ is separable but $l(x)^2$ is not. Contradiction.

Case 2: For any root ζ_n^m of $f(x)$ and any $p \nmid n$, ζ_n^{mp} is also a root of $f(x)$. In this case, we claim that $f(x) = \Phi_n(x)$. Indeed: if for any t that is coprime to n , there exists k coprime to n such that $t \equiv mk \pmod{n}$. (both t and m have inverse) Therefore, $\zeta_n^t = \zeta_n^{mk} = \zeta_n^{mp_1 p_2 \cdots p_i} \cdot p_i$ where p_1 to p_i are all coprime to n . Thus, ζ_n^t is a root of $f(x)$.

□

6. Thus, we see that $\mathbb{Q}(\zeta_n)$ is the splitting field of $\Phi_n(x)$. Since $\Phi_n(x)$ is irreducible and Galois, $\mathbb{Q}(\zeta_n)$ must be Galois. And its Galois group is $(\mathbb{Z}/n\mathbb{Z})^\times$

3 get to know it

1. Find $\Phi_n(x)$ for $N \leq 8$.
2. Find $\Phi_{2^m}(x)$
3. If n is odd, prove $\Phi_{2n}(x) = \Phi_n(-x)$
4. In general,

$$\Phi_{pn}(x) = \begin{cases} \Phi_n(x^p) & p \mid n \\ \frac{\Phi_n(x^p)}{\Phi_n(x)} & p \nmid n \end{cases}$$

5. Prove: integer $a > 1$, if $\Phi_n(a) \mid a - 1$, then $n = 1$.
In fact, $|x - 1|^{\phi(n)} < \Phi_n(x) < |x + 1|^{\phi(n)}$ for all x .
6. Prove: $\Phi_n(x) \mid \frac{x^n - 1}{x^d - 1}$ in $\mathbb{Z}[x]$ for all proper divisors d of n .

4 Wedderburn's Little Theorem

Theorem: Every finite division ring is a field.

1. If D is a finite division ring, then its center is a finite field. $Z = \mathbb{F}_q$ for some q .
2. D is a vector space over $\mathbb{F}_q$ with dimension n . So $|D| = q^n$.
3. Any centralizer is also a vector space, thus $|C_D(a)| = q^d$ for some $d \mid n$.

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4. Remove the 0 and look at them as a group. By class formula:

$$q^n - 1 = |Z| + \sum \frac{|G|}{|Stab_G(r)|} = q - 1 + \sum \frac{q^n - 1}{q^d - 1}.$$

5. So $\Phi_n(x) \mid (q - 1)$. So $n = 1$ and $D = Z$ is a field.

5 $\Phi_n(x)$ and order of an element mod p

1. Let a be an integer. If $p \mid \Phi_n(x)$, then $p \mid a^n - 1$
2. Moreover, if $p \nmid n$, $p \nmid a^d - 1$ for all $d < n$. Why? (separability comes into play here)
3. Thus, if $p \nmid n$, then $p \mid \Phi_n(a)$ iff the order of a in mod p is n .
4. Furthermore, $n \mid p - 1$ Why? Order of a divides $p - 1$!
5. Thus, there are infinitely many primes $\equiv 1 \pmod{n}$.

6 Zsigmondy's Theorem

1. Our goal is to find a prime such that $p \mid a^n - 1$ but not $a^d - 1$ for any $d < n$. So p must divide $\Phi_n(a)$.

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2. If we have a prime $p \mid \Phi_n(a)$, then we wish $p \nmid n$ to ensure that $p \nmid a^d - 1$ for all $d < n$. (separability consideration)
 3. So given a , is there a prime such that $p \mid \Phi_n(a)$ but $p \nmid n$?
 4. Mostly, yes.
 5. Assume not. Let $p_1, p_2, \dots, p_k$ be all the prime divisors of n . Then $\Phi_n(a)$ is consisted of $p_1, p_2, \dots, p_k$.
 6. By lifting the exponent lemma, we have $v_{p_i}(\Phi_n(a)) \leq v_{p_i}\left(\frac{a^n - 1}{a^{n/p_i^e} - 1}\right) + v_{p_i}(n) \leq v_{p_i}(n)$. So $(a - 1)^{\phi(n)} < \Phi_n(a) \leq n$. This is clearly wrong when a is large.